

Revision(15) **6015**

Signature

**MODEL Q.P**

Register No.

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/MANAGEMENT  
/COMMERCIAL PRACTICE**

**ELECTIVE II- CONCRETE TECHNOLOGY**

Time 3 hrs

Max marks 100

(I.S 456/2000, I.S 10262/2007 are permitted in the examination for reference)

**PART A**

(Answer all questions, each question carries 2 marks)

**I**

- 1 What are the functions of alumina in cement ?
- 2 Define bulking of sand?
- 3 Define creep in concrete.
- 4 Define concrete mix design
- 5 What is carbonation ?

(5x2=10 marks)

**PART B**

(Answer any five, each questions carries 6 marks)

- II.**
- 1.Explain hydration and what are the compounds formed during hydration
  2. What are the functions of chemical admixtures in concrete?
  3. List the terms associated with workability and discuss the factors affecting workability
  4. State the principal properties of hardened concrete
  5. State the objectives and basic principles of mix design
  6. How geo- polymer concrete is preparing
  7. List any six precautions to be taken in under water concreting

(5x6=30 marks)

## PART C

Answer one full question from each module, each question carries 15 marks

### MODULE I

III. a) What are the precautions taken for storing cement? Explain (8)

b) State the functions of ingredients in concrete with their limitations (7)

OR

IV. a) Classify different types of cements? (8)

b) Explain the factors which affect the alkali aggregate reaction (7)

### MODULE II

V. a) Explain the aggregate cement ratio and affects of cement content on the strength of concrete? (8)

b) Explain the importance of curing and what are the methods adopted for curing ? (7)

OR

VI. a) List the tests for measuring workability and explain slump test ? (8)

b) List the advantages of cement concrete? (7)

### MODULE III

VII. List the basic design requirements as per I.S 10262/2007 for mix design? (8)

b) State the purposes and requirements of concrete mix design? (7)

OR

VIII. Design a concrete mix of grade M30 with the following requirements

|                                                     |                        |
|-----------------------------------------------------|------------------------|
| Max size of aggr                                    | 20 mm                  |
| Degree of workability -                             | 0.90 compacting factor |
| Degree of quality control                           | good                   |
| Type of exposure                                    | moderate               |
| Cement OPC 43 grade, sp gr. 3.05,                   |                        |
| Sp. Gr. Of coarse aggregate 2., fine aggregate 2.60 |                        |
| Water absorption coarse 0.60%, fine 1.2%            |                        |
| River sand conforming to zone II                    |                        |

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### MODULE-IV

IX. a) Explain the test for self compactability of SCC? (8)

b) State the recommended practices and preservation of cold weather concrete? (7)

OR

X. a) State and explain various defects in concrete (8)

b) List the causes of cracks formation in concrete. (7)

(15x4=60 marks)